

EE 526 : Machine Learning

Mid-Semester Examination

Duration: 2 hours

Date: Feb 28' 2022

Marks: 50

Please outline all the steps systematically while presenting the solution.

1. It is known that the training samples $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ of a class ω_1 come from a two dimensional Gaussian probability density function of the form:

$$p(x_1, x_2) = K e^{-\frac{8}{7}(x_1^2 + \frac{3}{2}x_1x_2 + x_2^2)}$$

- (a) Find the value of the normalizing constant K that ensures that this function does satisfy the condition of being a probability density function.
- (b) Compute the mean vector and covariance matrix corresponding to this distribution.
- (c) Suppose the training samples are transformed using the matrix $\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$, compute the mean vector and covariance matrix corresponding to the transformed distribution.

The normal distribution $N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ in d -dimensions is given by :

$$p(\mathbf{x}) = \frac{1}{(2\pi)^{d/2} |\boldsymbol{\Sigma}|^{0.5}} e^{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})}$$

[3 + 2 + 3 marks]

2. (a) Suppose that we are given a sequence of training samples $\{x_1, x_2, \dots, x_N\}$, that are generated from a univariate normal distribution $N(\mu, 1)$. It is known that our prior knowledge on the mean μ follows a normal distribution $N(3, 2)$. Compute the MAP estimate of the mean μ . The normal distribution $N(\mu, \sigma^2)$ is given by

$$p(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$$

- (b) Compute the mode of the distribution

$$p(x|\omega_i) = \frac{x}{\theta^2} \exp^{\frac{-x^2}{2\theta-2}} \quad x \geq 0$$

- (c) It is to desired to apply a suitable transformation $\mathbf{A}$ on the training samples coming from a two dimensional Normal distribution of zero mean and covariance matrix $\begin{bmatrix} 3 & -1 \\ -1 & 3 \end{bmatrix}$, so as to make them independent. Find the matrix $\mathbf{A}$ that can achieve that.

[3 + 2 + 3 marks]

3. (a) In a signature verification system, there are two categories : forgery ω_1 and genuine ω_2 . The idea is to select a threshold, so that samples with scores less than this value can be regarded as being verified as genuine. It is known that the scores of the genuine and forgery follow a uniform distribution $U(4, 7)$ and $U(5, 10)$ respectively. Assuming a decision threshold of 6 and prior $P(\omega_1) = 0.6$, with zero one loss function, compute the:
- (i) average probability of a genuine signature getting rejected as a forgery signature.
 - (ii) average probability of a forgery signature getting falsely accepted as a genuine signature.
 - (iii) average probability of a signature to be accepted as a genuine.
 - (iv) average probability of a signature to be **correctly** accepted as a genuine.
 - (v) average probability of a signature to be rejected as a forgery.
 - (vi) average probability of a signature to be **correctly** rejected as a forgery.

[9 marks]

4. In a two class, two dimensional problem, the feature vectors in each class are normally distributed with identical covariance matrix : $\Sigma = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$. The respective mean vectors μ_1 and μ_2 (for class ω_1 and ω_2) are $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 3 \\ 1 \end{bmatrix}$. Assuming that the classes ω_1 and ω_2 are probable with 0.6 and 0.4, classify the feature vector $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$, according to the Bayes minimum probability classifier. [4 marks]

The normal distribution $N(\mu, \Sigma)$ in d -dimensions is given by :

$$p(\mathbf{x}) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{0.5}} e^{-\frac{1}{2}(\mathbf{x}-\mu)^T \Sigma^{-1}(\mathbf{x}-\mu)}$$

5. Consider a category ω_1 having a Geometric distribution $p(1-p)^{x-1}$, $0 \leq p \leq 1$ as the class conditional density, with parameter p . Note that x is discrete random variable
- (a) Given the set of N one dimensional training data $\{x_1, x_2, \dots, x_N\}$, estimate the ML for the parameter p .
 - (b) In the context of the Bayesian estimate problem, suggest a suitable expression of the conjugate prior.
 - (c) Compute the MAP estimate of the parameter p , given the form of the conjugate prior used in the above (b).

[3+ 2 + 3 marks]

6. (a) Consider two matrices $\mathbf{A} = \mathbf{X}\mathbf{X}^T$ and $\mathbf{B} = \mathbf{X}^T\mathbf{X}$, where $\mathbf{X}$ is of size $m \times n$ ($m \geq n$). Derive a relationship between the eigenvalues and eigenvectors of $\mathbf{A}$ and $\mathbf{B}$.
- (b) You are given 2 training examples (comprising 2 dimensions) for classes ω_1 and ω_2 .

$$\omega_1 : \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 2 \end{bmatrix} \quad \omega_2 : \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

Compute the optimal weight vector $\mathbf{w}$ (projection line in a reduced dimension) using the Fisher criterion.

If necessary, you may consider using the identity: $\begin{bmatrix} 2.5 & 0.5 \\ 0.5 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 0.44 & -0.22 \\ -0.22 & 1.1 \end{bmatrix}$.

[3+4 marks]

7. Consider a data set $\{\mathbf{x}_i, y_i\}_{i=1}^N$, where each input $\mathbf{x}_i$ is d -dimensional and its corresponding target y_i is an integer valued scalar, taking values +1 or -1. Additionally, assume that each data pair $(\mathbf{x}_i, y_i)$ is associated with a **known** weight r_i , (where $r_i > 0$). Accordingly, the sum of squares error function can be written as $E(\mathbf{w}) = \sum_{i=1}^N r_i(y_i - \mathbf{w}^T \mathbf{x}_i)^2$. Find the weight $\mathbf{w}^*$ that minimizes this function, using the least squares solution.

[6 marks]